u_sound

tetris_sound

audio

buzz

clear_evt

clk

frame_tick garb_evt hit_evt music_on

playing

sent_evt sfx_on
task_fail_evt
task_ok_evt
task_start_evt

clear_n[2:0]

pcm[15:0]

result[2:0]

FX_FRAMES[0,0]_i RTL_ROM

A[3:0] O[3:0]

FX_FRAMES[0,0]_i__0 RTL_ROM

A[3:0] O[3:0] FX_FRAMES[0,0]_i__1

RTL_MUX

I0[3:0] S=2'b00
I1[4:0] S=2'b01
I2[3:0] S=2'b10

I3[4:0] S=2'b11 O[4:0] S[1:0]

FX_FRAMES[0,0]1_i +RTL_ADD

I1

I0[1:0] O[1:0]

FX_FRAMES[0,1]_i RTL_ROM

A[3:0] O[4:0] FX_FRAMES[0,2]_i RTL_ROM
A[3:0] O[3:0] FX_FRAMES[0,3]_i RTL_ROM

A[3:0] O[4:0]

FX_NOTE[0,0]_i RTL_ROM

A[3:0] O[4:0]

FX_NOTE[0,0]_i__0 RTL_ROM

A[3:0] O[4:0]

FX_NOTE[0,0]_i__1

RTL_MUX

I0[4:0] S=2'b00
I1[4:0] S=2'b01
I2[4:0] S=2'b10

I3[4:0] S=2'b11 O[4:0] S[1:0]

FX_NOTE[0,1]_i RTL_ROM

A[3:0] O[4:0]

FX_NOTE[0,2]_i RTL_ROM

A[3:0] O[4:0] FX_NOTE[0,3]_i RTL_ROM

A[3:0] O[4:0]

bucket_reg[12:0] RTL_REG CD Q

bucket0_i +RTL_ADD

I0[12:0]

I1[11:0] O[12:0] buzz_r_reg RTL_REG CD Q

fx_id_reg[3:0]

RTL_REG C

CED Q

fx_id0_i >=RTL_GEQ O

I0[4:0]
I1[4:0]

fx_left_i RTL_MUX S

I0[5:0] S=1'b1

I1[5:0] S=default O[5:0] fx_left_i__0 RTL_MUX S

I0[3:0] S=1'b1

I1[5:0] S=default O[5:0]

fx_left_i__1 RTL_MUX

I0 S=1'b1

I1 S=default O S

fx_left_i__2 RTL_MUX

I0 S=1'b1

I1 S=default O S

fx_left_i__3 RTL_MUX

I0 S=1'b1

I1 S=default O S

fx_left_i__4 RTL_MUX

I0 S=1'b1

I1 S=default O S

fx_left_reg[5:0]

RTL_REG C

CED Q -

fx_left0_i RTL_SUB

I0[5:0]

I1[5:0] V=B"000001" O[5:0]

fx_now_i RTL_MUX S

I0[4:0] S=1'b1

I1[4:0] S=default O[4:0]

fx_now_i__0 RTL_MUX

I0 S=1'b1

I1 S=default O S fx_now_i__1 RTL_MUX

I0 S=1'b1

I1 S=default O S

fx_now_i__2 RTL_MUX

I0 S=1'b1

I1 S=default O S

fx_now_i__3 RTL_MUX

I0 S=1'b1

I1 S=default O S fx_now_i__4 RTL_MUX

I0 S=1'b1

I1 S=default O S

fx_now_i__5 RTL_MUX

I0 S=1'b1

I1 S=default O S

fx_now_reg[4:0]

RTL_REG_SYNC C

CED Q

RST

fx_step_i RTL_MUX

I0 S=1'b1

I1 S=default O S

fx_step_i__0 RTL_MUX

I0 S=1'b1

I1 S=default O S
fx_step_i__1 RTL_MUX S

I0[2:0] V=B"011", S=1'b1

I1[2:0] S=default O[2:0]

fx_step_i__2 RTL_MUX S

I0[2:0] S=1'b1

I1[2:0] S=default O[2:0]

fx_step_i__3 RTL_MUX S

I0[2:0] S=1'b1

I1[2:0] S=default O[2:0] fx_step_i__4 RTL_MUX S

I0[2:0] S=1'b1

I1[2:0] S=default O[2:0]

fx_step_i__5 RTL_MUX S

I0[2:0] V=B"100", S=1'b1

I1[2:0] S=default O[2:0]

fx_step_i__6 RTL_MUX S

I0[2:0] S=1'b1

I1[2:0] S=default O[2:0] fx_step_i__7 RTL_MUX S

I0[2:0] S=1'b1

I1[2:0] S=default O[2:0]

fx_step_reg[2:0]

RTL_REG_SYNC C

CED Q

RST SET

fx_step0_i +RTL_ADD

I1

I0[2:0] O[2:0]

fx_step1_i RTL_OR

I0

I1 O

fx_step1_i__0 RTL_AND

I0

I1 O

fx_step1_i__1 >RTL_GT O

I0[5:0]
I1[5:0] V=B"000001"

fx_step2_i =RTL_EQO

I0[2:0]
I1[2:0] V=B"011"

fx_step2_i__0 =RTL_EQO

I0[4:0]
I1[4:0]

fx_step2_i__1 <RTL_LT O

I0[2:0]
I1[2:0] V=B"100"

half_period_i RTL_ROM

A[4:0] O[18:0]

loudness_i RTL_MUX S

I0[11:0] S=1'b1

I1[11:0] V=B"011000000000", S=default O[11:0]

loudness_i__0 RTL_MUX S

I0[11:0] S=1'b1

I1[11:0] S=default O[11:0]

loudness0_i +RTL_ADD

I0[11:0] V=B"011000000000"

I1[11:0] O[11:0]

loudness0_i__0 +RTL_ADD

I0[11:0]

I1[10:0] V=B"10000000000" O[11:0]

loudness1_i RTL_AND

I0

I1 O

playing_d_reg RTL_REG CD Q

result_d_reg[2:0] RTL_REG CD Q

s_cnt_i RTL_MUX

I0 S=5'b00000

I1 S=default O

S[4:0]

s_cnt_reg[18:0] RTL_REG_SYNC CD Q

RST

s_cnt0_i +RTL_ADD

I1

I0[18:0] O[18:0]

s_cnt0_i__0 >=RTL_GEQ O

I0[19:0]

I1[19:0] -s_cnt1_i RTL_SUB

I0[18:0]

I1[30:0] V=X"00000001" O[19:0] s_sq_i RTL_ROMO A[4:0] s_sq_reg RTL_REG_SYNC C

CED Q

RST

s_sq0_i RTL_INV

I0 O

start_fx_i RTL_MUX S

I0[1:0] V=B"01", S=1'b1

I1[1:0] S=default O[1:0]

start_fx_i__0 RTL_MUX

I0[2:0] V=B"010", S=3'b100

I1[2:0] S=default O[2:0] S[2:0]

start_fx_i__1 RTL_MUX S

I0[3:0] S=1'b1

I1[3:0] S=default O[3:0]

start_fx_i__2 RTL_MUX S

I0[4:0] V=B"00011", S=1'b1

I1[4:0] S=default O[4:0]

start_fx_i__3 RTL_MUX S

I0[4:0] V=B"00110", S=1'b1

I1[4:0] S=default O[4:0]

start_fx_i__4 RTL_MUX S

I0[4:0] V=B"00111", S=1'b1

I1[4:0] S=default O[4:0] start_fx_i__5 RTL_MUX S

I0[4:0] V=B"01001", S=1'b1

I1[4:0] S=default O[4:0] start_fx_i__6 RTL_MUX S

I0[4:0] V=B"01000", S=1'b1

I1[4:0] S=default O[4:0] start_fx_i__7 RTL_MUX S

I0[4:0] V=B"01010", S=1'b1

I1[4:0] S=default O[4:0] start_fx_i__8 RTL_MUX S

I0[4:0] V=B"01011", S=1'b1

I1[4:0] S=default O[4:0]

start_fx_i__9 RTL_MUX S

I0[4:0] V=B"00101", S=1'b1

I1[4:0] S=default O[4:0] start_fx_i__10 RTL_MUX

I0[4:0] V=B"00100", S=3'b001

I1[4:0] S=default O[4:0] S[2:0]

start_fx_i__11 RTL_MUX S

I0[4:0] S=1'b1

I1[4:0] S=default O[4:0]

start_fx1_i >RTL_GT O

I0[2:0]
I1[2:0]

start_fx1_i__0 RTL_AND

I0

I1 O

start_fx1_i__1 >=RTL_GEQ O

I0[2:0]
I1[2:0] V=B"010"

start_fx1_i__2

!=RTL_NEQ O

I0[2:0]

I1[2:0] start_fx2_i =RTL_EQ

I0

I1 O